

Design and implementation of pneumatic actuators in a dual-axis solar tracker under the climatic conditions of karbala city

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ABSTRACT

The inability of stationary solar panels to track the sunlight is one of its primary drawbacks. It will produce less electrical energy as a result. This issue is resolved by combining these panels with appropriate actuators to create solar trackers. Solar panels can continuously monitor the sun's rays thanks to the actuators, enabling them to generate maximum energy during the day. Pneumatic actuators were used because of their advantages over others. This study aims to provide a simple and comprehensive design approach for pneumatic actuators that relies on factors including inclination angle, actuator displacement, and required power to drive a closed-loop dual-axis solar tracker adapted to the harshest climate in Karbala city. Three consecutive days were allocated to conduct experimental testing of the system. According to the findings, the average efficiency of the solar panel was raised by 62.3 % when it was moved on one axis using a single actuator and by 66.4 % when it was moved on two axes using two actuators. A stability investigation of the pneumatic cylinder movement under PID control unit revealed that the system reacts satisfactorily, with steady-state error of 0.00734 %, overshoot of 0.505 %, settling time of 1.034 s, and rising time of 145.76 ms.

1. Introduction

Today, sustainability is becoming more and more important. Users are beginning to favor renewable energy sources [1] over fossil fuels, such as solar, wind, and other energy sources. The appeal of solar energy as a fossil fuel substitute is growing. Solar energy may be used for many different things, such as air heating systems [2,3], water heating systems [4,5], agricultural drying systems [3,6], producing electricity [7,8], and more. Numerous techniques, including hydro, solar, wind, and ecological energy, are used to meet the increasing need for power and achieve sustainability goals [7].

Solar energy equipment is in more demand due to the growing importance of clean energy. Solar-powered devices are widely used in residential, business, and agricultural contexts. Powered by a solar tracking system, this equipment's most noticeable feature is its solar panel, which self-aligns with the sun's position to capture solar energy [9]. These devices need the installation of solar tracker actuators, which allow the solar panels to be oriented as best they can toward the sun, to guarantee that these panels get sunlight throughout the day.

Since actuators are crucial components of solar tracking systems,

their design requires careful consideration. Actuators must be exceptionally durable, stable, precise, technically controlled, and able to work seamlessly with solar trackers.

Solar trackers are devices that monitor the sun's path across the sky to maximize the amount of sunlight that reaches PV cells and boost their power output. In comparison to stationary solar panels, solar panel efficiency may be boosted by 30–40 % with the use of a solar tracking system combined with a linear actuator or geared motor [9]. Sensors, controllers, and rotary or linear actuators are commonly used in solar trackers [10] to enable good sun tracking.

Actuators can be energized by electrical power, hydraulic pressure, or pneumatic pressure. They change energy into motion. They enable users to change the angle of the PV panel. The two basic types of solar tracking systems are single-axis [11,12] and dual-axis [13,14]. The first type usually uses the azimuth angle to track the sun on one axis, whereas the latter uses two axes to track the sun, one for each of the azimuth and altitude solar angles. Recently, new designs and technologies for both types of tracking systems have been created to provide high structural dependability and control precision for the practical performance of these systems.

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There are many types of actuators, such as linear motors [15,16], servo motors [17,18], DC motors and gearboxes [19,20], and stepper motors [12,21]. To drive the solar tracker, additional actuators including thermo-mechanical [22,23], hydraulic [24,25], and pneumatic [26,27] ones can be employed.

Pneumatic actuators are tools that transform compressed air energy into mechanical energy for productive work. In other words, they work to generate displacement through the movement of the piston or to produce the necessary force at the stroke end.

In many situations, pneumatic actuators are preferred over electric and hydraulic actuators due to their many benefits [28]. They are less costly, have longer operating lifespans, require less maintenance, and are simpler to operate than electric actuators. Pneumatic systems may be developed and used more easily than electrical ones for many applications. The production costs and contamination resistance of pneumatic actuators are lower than those of hydraulic actuators. Pneumatic actuator systems are also not much impacted by changes in external temperature. Since collecting actuator exhaust gases is not required, fluid tubes are not required. With the pneumatic systems almost completely dry, long-term storage is not required. Compared to hydraulic controllers, they are also more reliable, simpler to operate, cleaner, more readily accessible, safer for the environment, and safer for employees.

In general, pneumatic actuators have some benefits over other actuators [28].

- Less expensive and much simpler design.
- Offer an excellent force-to-weight ratio.
- Exhibit better force control and superior positioning control.
- Improved precision.
- Better safety features.
- Better for the environment.
- A wider range of applications
- More reliable than hydraulic controllers

Pneumatic actuators, often known as artificial muscles, are used in the automotive industry [29], robotics and automation [30], medical devices [31], lifting systems [32], tracking systems [26], and other fields.

Pneumatic actuators with sufficient power and efficiency can be used with solar panel tracking systems [26,27,33] to create smoothly controlled motion. For solar trackers, pneumatic actuators provide an inexpensive means of height adjustment. The low-duty cycle functioning of the pneumatic tracker allows for future monitoring and scaling while requiring less maintenance and energy savings. Flexibility, outstanding durability, reliability, and responsiveness should all be considered in the design of the solar tracker's actuators.

Below is a review of some studies concerning solar trackers powered by electric motors or hydraulic actuators. A new algorithmic method for a semi-continuous single-axis solar tracking system driven by a linear actuator was presented by Azam et al. [34]. The tracker output power has increased due to improved solar radiation collection and reduced self-energy use. The semi-continuous tracker was shown to use around 34 % less energy than a continuous tracking system. To optimize the power point tracking performance of a dual-axis solar tracker powered by a DC motor with a gear reducer, Solís-Cervantes et al. [35] devised and constructed a controller governed by sensorless feedback and dynamics of a classical proportional integration scheme. The results showed that the movable tracker produced 27.75 % more power than the fixed one. Kumar et al. [36] experimentally investigated a two-axis-GPS sun tracker directed by gear DC motors. They proved that this tracker outperformed the static based- and LDR-based PV systems by 45.85 & 14.93 %, respectively. Song et al. [37] demonstrated a comparison between a fixed PV panel system and a PV panel-based passive solar tracking system driven by a worm gear reduction motor. The tracker could increase the energy generation efficiency of solar cells by 28.5–42.9 % during the spring season in Qingdao, East China. The

tracker may boost PV energy production by 37 % on partly cloudy days.

Rotar et al. [38] demonstrated a self-sufficient dual-axis solar tracking system using a stepper motor to regulate PV panel height and azimuth angles. A new set of maintainability-oriented metrics was proposed to track device evaluation and improvement to create a maintainability index. This index enhanced the repairability and maintainability of the system and increased the uptime. P.-Jara et al. [39] studied solar energy efficiency in the coastal city of Manta by comparing fixed PV systems with two single-axis solar trackers driven by a stepper motor, one based on astronomical programming and the other based on LDRs. The LDR-based astronomical trackers have been shown to outperform a fixed PV system with net gains of 31.8 % and 37 %, respectively.

Fresnel lens was added by Ayu et al. [40] to enhance the efficiency of a dual-axis solar tracker. NodeMCU ESP32, Arduino Nano, Fresnel lens, servo motors, and other sensors were combined with the tracker. The results showed that the tracker's maximum power with and without the lens was 13.60 W and 15.78 W, respectively. Wu et al. [41] implemented a dual-axis solar tracker with a satellite compass and inclinometer for automatic positioning and tracking. The tracker was positioned by servo motors under open-loop control with GPS. The tracker extracted more electricity than the fixed-tilted tracker by 35.91 % in fixed locations with largely clear days, 38.72 % and 38.4 % in varied locations with clear days, as well as, 60 % and 113.7 % in heading variation and movement circumstances, respectively.

A solar tracking system that uses hydraulic actuators for tracking was devised and developed by Tambe et al. [42]. This is the first attempt to provide a viable tracking system for remote locations and to use gravity energy as a driving force for solar tracking devices. Tracking using mechanical energy will boost solar panel production and eliminate the tracking system's position restriction. The tracker produced up to 26.9 % more power than a fixed-tilted PV system in clear conditions. Ayubova et al. [43] developed novel circuit solutions for the solar track hydraulic drive that regulate the drive's electro-hydraulic distributors according to data coming from the sun position sensor and hydraulic actuator sensors. The fundamental hydraulic actuator has been mathematically modeled for circuit tracking, considering the variety of influencing parameters to maximize the hydraulic system performance. The outcomes validated the control system's accuracy and efficiency, establishing the hydraulic actuator as a crucial component in the creation of sustainable and effective energy. Hanfesh et al. [25] created and put into use a solar tracker that uses a hydraulic system with two hydraulic cylinders to steer a 160W PV panel in two directions. Extremely simple control and ease of mobility were characteristics of the system. An Arduino microcontroller-developed algorithm determined the path's position. The results showed an average power gain of 18 % above the fixed panel system, the solar tracking system has demonstrated superior performance.

Few studies in the literature have addressed the topic of solar trackers driven by pneumatic actuators. Baisrum et al. [35] employed a pneumatic cylinder in the solar tracker design. An ultraviolet (UV) sensor was employed to track the movement's angle. The error values derived from the outcomes of processing UV sensor data are fed into fuzzy control-based decision-making systems. The pneumatic cylinder's position can travel up, down, or halt depending on the decision-making system's output. The test apparatus functioned as intended, with a 5.3 % dynamic response overshoot and a 1.6 % steady-state inaccuracy. Baisrum et al. [44] designed and implemented a prototype of a hybrid system integrated with pneumatic actuators and electronic control systems. The tracker was controlled by ANFIS approach. The combined reading feedback of two UV sensors positioned on the east and west sides of the solar panel provided the ANFIS input. According to the experimental findings, the average inaccuracy in the east-west direction was 1.6° for no-load situations and 2.5° for loaded ones. 2.08 % overshoot and 1.25 % steady-state inaccuracy were the results of the system's dynamic reaction. Fathi et al. [27] suggested an experimental pneumatic

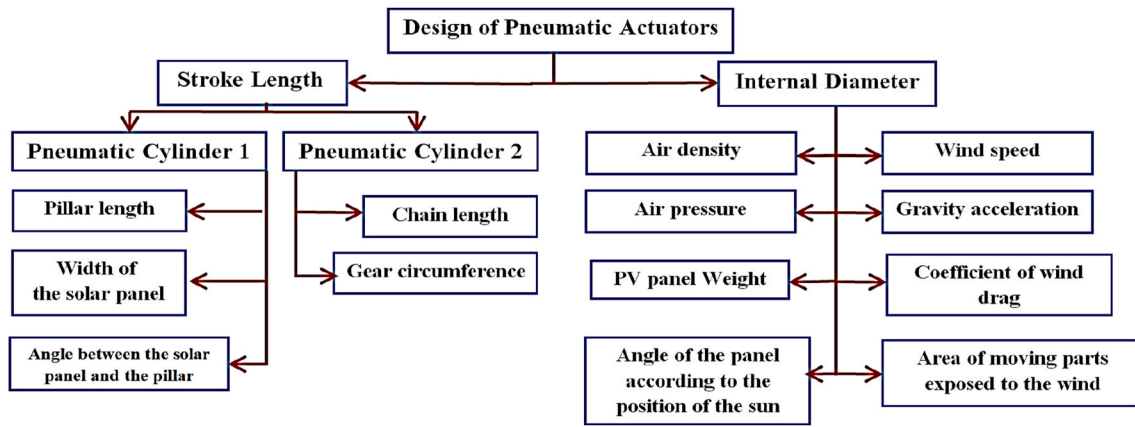


Fig. 1. Pneumatic mechanism design steps.

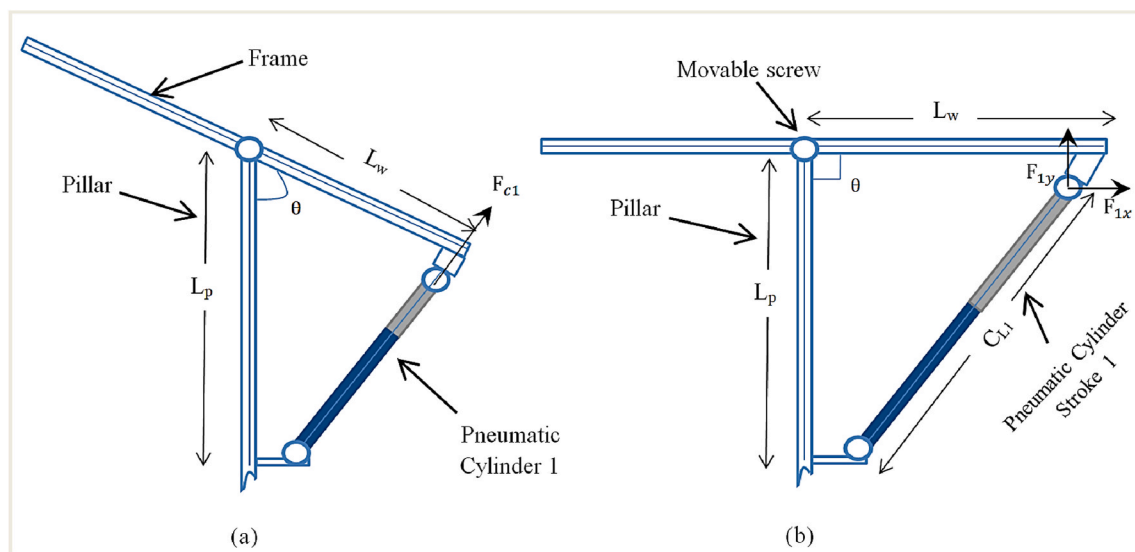


Fig. 2. The pneumatic cylinder 1 with (a) min. Returning and (b) max. extension.

dual-axis sun tracker employing light dependent resistor (LDR) sensors. The pneumatic tracker's low-duty-cycle operation, which results in monitoring energy conservation, reduced maintenance requirements, and the possibility for scaling, is made possible by its inbuilt energy storage. The real supplied power of the solar PV is computed for the first time, taking into account the expenses of solar tracking and maximum power point tracking (MPPT) power, even if MPPT guarantees maximum load power supply. According to the testing findings, the suggested solar tracking system outperformed the fixed panel system by 37.6 % overall and by 35.3 % in terms of real power output, with an MPPT efficiency of 90 %. Under fuzzy logic control (FLC), Khalaf et al. [45] used Matlab/Simulink to construct a dual-axis solar tracker powered by electro-pneumatic actuators to maximize solar radiation employing LDR sensors. In this system, the FLC and LDR values are regarded as inputs that regulate the solar panel's orientation to achieve the optimal tilt and azimuth angles. Born et al. [26] integrated two actuators into a compliant glass fiber-reinforced plastic (GFRP) elastomer-hybrid composite in the PV-functionalized FlectoSol module, enabling a deflection in two directions $\pm 80^\circ$, necessary for sun tracking.

The present study differs from every other solar tracking study previously described in the literature, as some used electric motors and others used hydraulic actuators to drive sun trackers. Relevant studies that used aero actuators are very few. In general, all studies did not include design steps for the actuators in the detail that the current study

adopted and under different climatic conditions. This study focuses on designing the pneumatic actuators needed to drive a closed-loop dual-axis solar tracker at the lowest costs, taking into account the maximum load under the worst climatic conditions in Karbala city, including wind speed and impact angles. This design was achieved by conducting comprehensive calculations based on the necessary equations. Two double-acting pneumatic cylinders will be used in the tracking system as actuators to track the sun's daily movements from east to west and seasonal movements from north to south. The inner piston diameter and cylinder stroke length are included in the design. Two valves and throttles were used to regulate the position of each pneumatic actuator to facilitate the movement of the cylinder. A rotating disk and chain on a slide converted the linear motion of one of the cylinders into rotational motion. The stability of the pneumatic system was also analyzed by subjecting it to a PID-based closed-loop control. This will have a positive impact on the performance of the solar tracking system.

2. Research Methodology

2.1. Design of the pneumatic cylinders

This study proposes selecting and designing pneumatic actuating mechanisms by adopting mathematical computations for two pneumatic actuators numbered "1" and "2" to develop the operation of a dual-axis

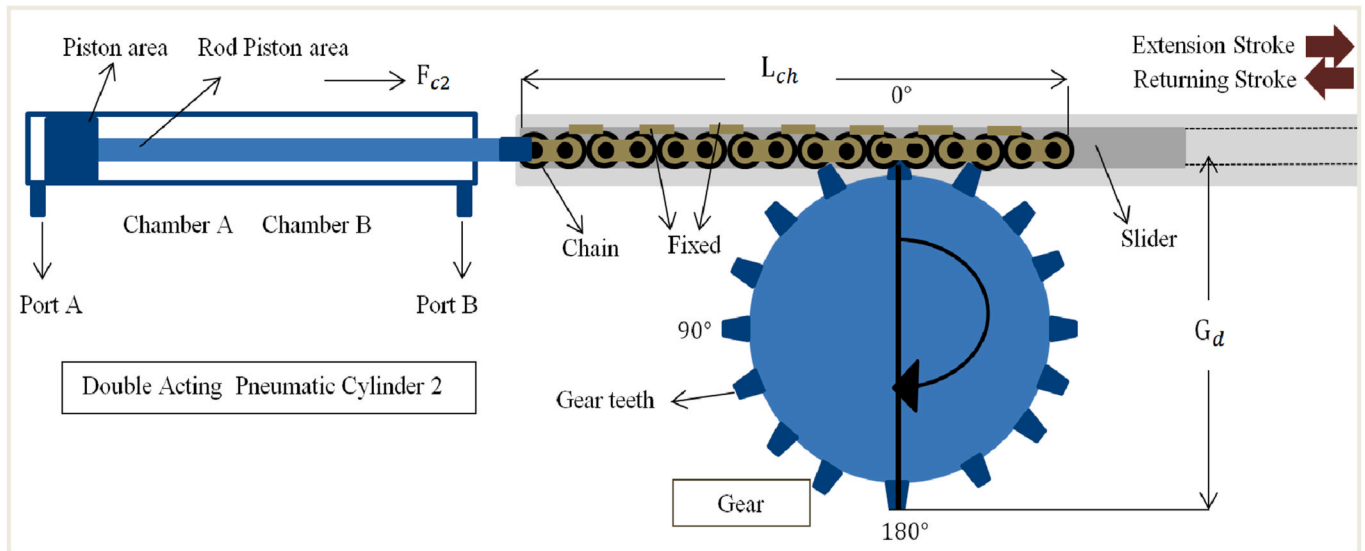


Fig. 3. The movement mechanism with pneumatic cylinder 2.

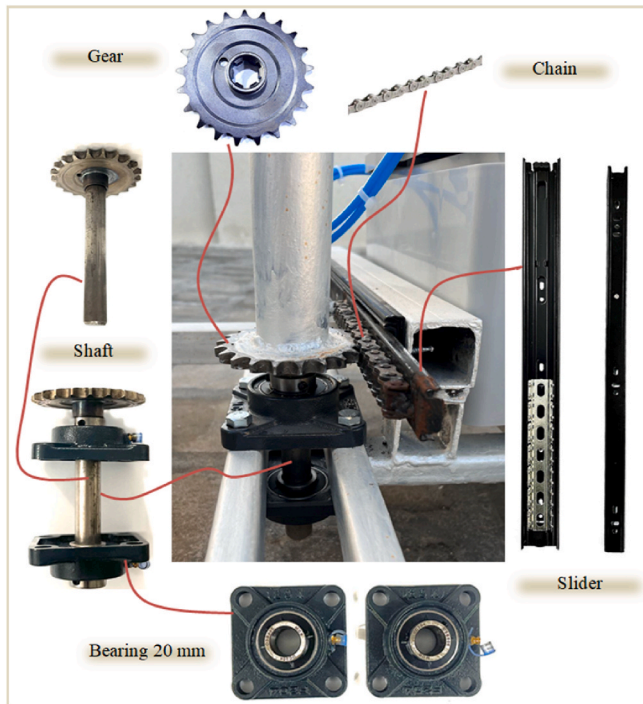


Fig. 4. Model for converting the linear motion of pneumatic actuator 2 into rotational.

solar tracker for a solar panel. Cylinder 1 is utilized to rotate the solar panel in the vertical direction from south to north, while cylinder 2 is used to rotate the panel from east to west. Fig. 1 illustrates the design steps for the pneumatic mechanism.

2.1.1. Stroke length design of the pneumatic cylinders

The Pythagorean law of lengths can be applied to the schematic shown in Fig. 2 to calculate the full length of cylinder 1 (C_{L1}) when extended to its end (cylinder length + maximum stroke).

$$C_{L1}^2 = L_w^2 + L_p^2 - 2 L_w L_p \cos(\theta) \quad (1)$$

Where $L_w = 212.5$ mm is half the width of the solar panel, $L_p = 430$ mm is half the pillar length, and θ is the angle between the solar panel

and the pillar. $\theta = 90^\circ$ when the cylinder extends to its maximum stroke. The south-north rotation axis of the tracking system will be aligned with the center axis of the frame. By substituting the values of all variables in Eq. (1), the value of C_{L1} is found to be equal to 480 mm (Represents the total length of fully extended cylinder 1 with top and bottom cylinder holders). The length of each holder at each end of the cylinder is 40 mm. There is also a 10 mm part of the cylinder rod that remains inside the chamber and does not contribute to the effective length of the rod. As a result, the net stroke length of pneumatic cylinder 1 to provide the required altitude angle is $(480/2) - (2 \times 40 + 10) = 150$ mm.

To design the required movement of pneumatic cylinder 2 shown in Fig. 3, a gear with a diameter G_d of 108 mm and a chain with a length L_{ch} of 180 mm were chosen. The chain must be designed so that it can rotate half the gear. The required chain length $L_{ch} = \pi G_d/2 \approx 170$ mm. 180 mm chain length was chosen for safe design purposes.

In this way, the gear teeth can be interlocked in the chain. Increasing the length of the chain does not affect the movement aspect because the main control of the stroke length is the cylinder whose length will be designed. The chain must also be mounted on a stabilizer that can move when pushed by cylinder 2, as shown in Fig. 4. The chain and gear contribute to converting the linear movement of cylinder 2 into the rotational motion required to direct the solar panel from east to west.

The pillar is welded over the gear. The slider on which the chain is fixed is used to stabilize the chain and drive the pneumatic cylinder smoothly during the process of converting the linear motion of cylinder 2 into a rotational motion. The slider must be installed with the base to hold the pillar controlling the movement accurately and preventing the stroke of the cylinder from going out of its specified path.

Gear circumference = $\pi \times G_d = \pi \times 108 \approx 340$ mm. To achieve rotation of the panel at 180° , only half a cycle is needed in the movement. So the circumference of the circle must be divided by two; $340/2 = 170$ mm. Therefore, the effective stroke length of cylinder piston 2 equals $170 - 20 = 150$ mm, when 20 mm represents the portion of the cylinder length that remains inside the chamber and is not included in the calculations of the active length of the stroke.

2.1.2. Internal diameter design of the pneumatic cylinders

Several factors must be taken into account when designing suitable actuators for a solar tracker. These factors are represented by the range of loads that the panel experiences when it rotates, some of which are internal, represented by the weight and size of the panel and its rotating accessories, and some are external, representing the wind load. The wind load affects how close the solar PV panels must be positioned to the

Table 1
Practical specifications of pneumatic cylinders.

Item	Quantity	Specification
Pneumatic cylinder	2	Double-acting cylinder, EXP- FLEX-MAC 30 × 155-S-0.05–1.0 Mpa with stroke of 155 mm. The force (Max load) cylinder output 350 N, cylinder return force 307 N, full retraction (320 mm) length & full extension (340 mm) length were chosen.

Table 2
Design dimensions of pneumatic actuators.

Dimensions (mm)	Pneumatic actuator 1	Pneumatic actuator 2
Cylinder stroke length	150	150
Cylinder internal diameter	22.71	25.5
Cylinder fully extended length	340	340
Cylinder fully retracted length	320	320
Chain length	–	180
Gear diameter	–	108

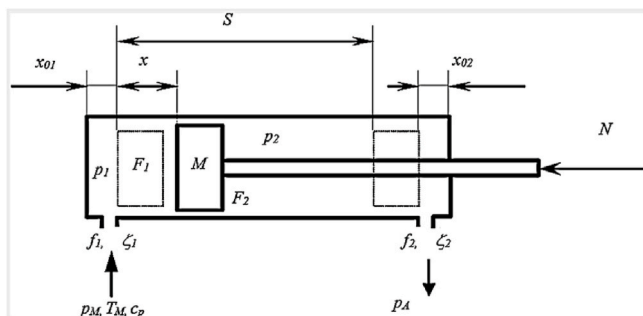


Fig. 5. Design diagram of a pneumatic actuator [47].

roof's edges and exerts excessive stress on the PV modules and PV mounting system, making it a significant consideration. Because friction load is so little, it is ignored in all computations. Heavy lateral loads require more planning because cylinders are primarily made to push and draw weights. Consideration should be given to designing the cylinder thrust to remain as near to the piston rod's centerline as feasible. When selecting the kind of cylinder and size of the piston, the main factor is the load.

Since the pneumatic cylinder 1 is in an inclined position relative to the level of the solar panel as shown in Fig. 2, the cylinder force must be decomposed into two components and entered into the equation.

The maximum wind force F_w facing the panel and the force component F_{1y} perpendicular to the panel added together and divided by $\sin \theta$ equals the total maximum force F_{c1} operating on cylinder 1:

$$F_{c1} = \frac{F_{1y} + F_w}{\sin \theta} \quad (2)$$

$$F_{c1} = \frac{1}{\sin \theta} \left[(M_1 g)_{int \ load} + \left(\frac{1}{2} \times \rho V^2 A_w C_d \right)_{wind \ load} \right] \quad (3)$$

Where F_c in N is the actuator's lifting force and θ in rad is the angle of the panel according to the position of the sun, $M_1 = 2$ kg is half the weight of the PV panel, since the panel weight is equally distributed on both sides of cylinder 1 around the midpoint of the frame, $C_d = 1.05$ [27] is the coefficient of wind drag, $A_w = 0.67 \times 0.425$ m² is the area of moving parts exposed to the wind, $V = 5.5$ m/s is the wind speed, $\rho = 1.29$ kg/m³ is the air density, and $g = 9.8$ m/s² is the acceleration due to gravity. The optimal lifting force is achieved by selecting θ at 45° [46]. After performing the necessary calculations on the previous equation, it was found that $F_{c1} \approx 36$ N.

Eq. (3) can be reformulated to calculate the force of cylinder 2 required to rotate the solar panel horizontally, with the aid of Fig. 3:

$$F_{c2} = [M_2 g]_{int\ load} + \left[\frac{1}{2} \times \rho V^2 A_w C_d \right]_{wind\ load} \quad (4)$$

Where $M_2 = 4 \text{ kg}$ is the weight of the solar panel. After performing the necessary calculations on the previous equation, it was found that $F_{c2} = 45 \text{ N}$.

The required diameter of pneumatic actuators can be calculated from the force equation:

$$F_c = P \times A \quad (5)$$

Where A is the area of the cylinder and P is the air pressure. Using the force value of $2 \text{ Bar} = 2 \times 10^5 \text{ N/m}^2 = 2 \times 10^5 \text{ Pa}$. Substituting the cylinder force and pressure values in Eq. (5), the cylinder area A_1 was found to be $1.8 \times 10^{-4} \text{ m}^2$ while the cylinder area $A_2 = 2.25 \times 10^{-4} \text{ m}^2$. Substituting the areas, A_1 and A_2 in Eq. (6) to find the cylinder diameter of cylinders:

$$A = \frac{\pi}{4} \times D^2 \quad (6)$$

Where D is the cylinder diameter. It is found that $D_1 = 15.14$ mm and $D_2 = 17$ mm. Using a safety factor of 1.5, the safe diameters of pneumatic cylinders 1 and 2 are 22.71 mm and 25.5 mm, respectively. The design of the pneumatic actuator might be aided by the actuator's size and force design. In this study, the pneumatic cylinders of the solar tracking system may be chosen for practical applications with the following specifications listed in [Table 1](#). All the design dimensions of the pneumatic cylinders and transmission mechanisms are listed in [Table 2](#).

2.2. Behavior of the pneumatic actuator under PID controller

2.2.1. Mathematical model of the pneumatic actuator

The unstable nature of compressed gas flows makes it difficult to model the operations of pneumatic systems. For this reason, when creating the actuator's mathematical model, the following assumptions were made [47].

- The compressed air supply's characteristic remains constant.
- The gas's state changes thermodynamically through an adiabatic process.
- Ideal gas model is utilized.
- The flow coefficients of the control devices remain constant.

The mathematical model [47] of the pneumatic actuator was created based on the diagram shown in Fig. 5.

The output link of the pneumatic actuator's equation of motion is:

$$M \frac{d^2 x}{dt^2} = p_1 f_1 - p_2 F_2 - p_A (F_1 - F_2) - N \quad (7)$$

The outlet space's change in air pressure p_1 is:

$$\frac{dp_1}{dt} = \frac{k f_1 \sqrt{R T_m}}{F_1(x + x_{o1})\sqrt{\zeta_1}} \sqrt{p_M^2 - p_1^2} - \frac{k p_1}{(x + x_{o1})} \frac{dx}{dt} \quad (8)$$

The outlet space's change in air pressure p_2 is:

$$\frac{dp_2}{dt} = -\frac{k f_2 \sqrt{R T_M}}{F_2(S-x-x_{o2})\sqrt{\epsilon_2}} \left(\frac{p_2}{p_M}\right)^{\frac{k-1}{2k}} \sqrt{P_2^2 - P_A^2} + \frac{k p_2}{(S-x+x_{o2})} \frac{dx}{dt} \quad (9)$$

As the pneumatic piston goes from stop to stop, the coordinate of its location, x , fluctuates between 0 and S . The mathematical model must take this constraint into account. It is necessary to immediately add the condition to the acquired equations of the pneumatic actuator's mathematical model to artificially set acceleration, velocity of motion, and coordinate equal to zero in the program if the pneumatic actuator is in

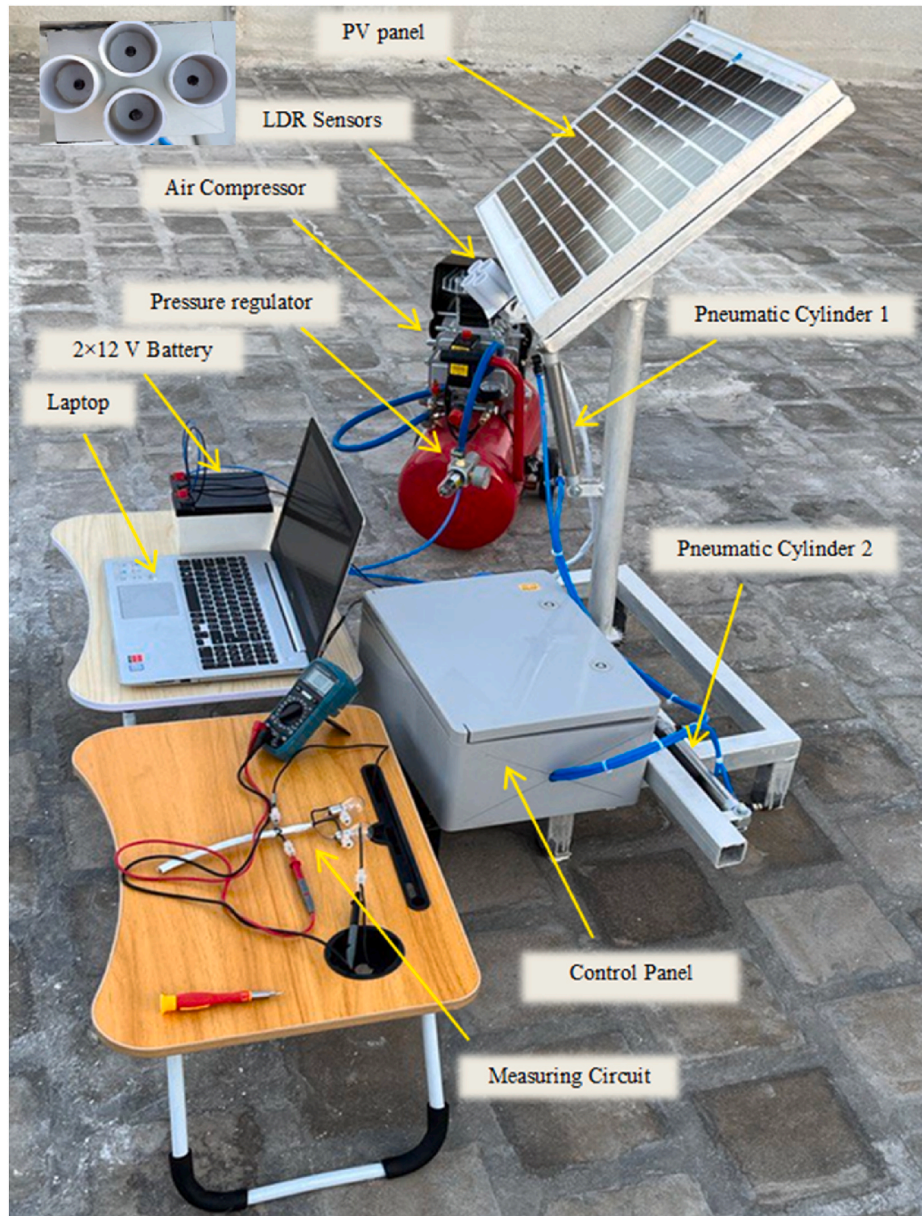


Fig. 6. The entire setup of the sun tracking system.

the leftmost position.

$$T_o \frac{d^2x}{dt^2} = \frac{dx}{dt} = x = 0 \quad \text{if } x < 0 \quad (10)$$

Likewise, if the piston ends up at the far right, a condition that restricts the x coordinates to the actuator stroke length S has to be written:

$$T_o \frac{d^2x}{dt^2} = \frac{dx}{dt} = x = 0 \quad \text{if } S \geq 0 \quad (11)$$

Where M is the piston, rod, and external load mass (kg), p_1 and p_2 are pressures in the head and rod ends (P_a), respectively, p_A is the outlet pressure (P_a), F is the piston effective area (m^2), N is the static load (N), Tm is the absolute gas temperature (K), k is the adiabatic index (Poisson's coefficient), p_M is the supply pressure (Pa), R is the gas constant (J/(kg K)), f is the pipe cross-section area (m^2), x is the piston position coordinate (m), x_{01} is the reduced coordinate (m) (describes the damaging head end of the piston cavity), x_{02} is the reduced coordinate (m) (describes the rod cavity's hazardous space volume), ζ is the line resistance factor, and S is the peak piston stroke (m).

2.2.2. Pneumatic actuator under PID controller

The study examines a technique for managing the movement of the tracking system's output links for pneumatic actuators of technical equipment. The PID controller improves the dynamic and qualitative features. The most prevalent regulator equation in servomechanisms and process control is [48]:

$$u(t) = u_o + K \left[e(t) + \frac{1}{T_i} \int_0^t e(\tau) d\tau + T_d \frac{de(t)}{dt} \right] = u_o + u_p(t) + u_i(t) + u_d(t) \quad (12)$$

The output signal generated by the PID controller is the total of the three elements of derivative, integral, and proportional control. The output quantity error, or the difference between the output quantity and the reference value, is proportional to the first component $u_p(t)$. The second part $u_i(t)$ is integral in time of the output quantity error, and the third part $u_d(t)$ is derivative of the error. Where u_o is the offset or correction value that modifies the controller's average output level, K is the controller gain, and T_i and T_d are the integration and differentiation time constants (s).

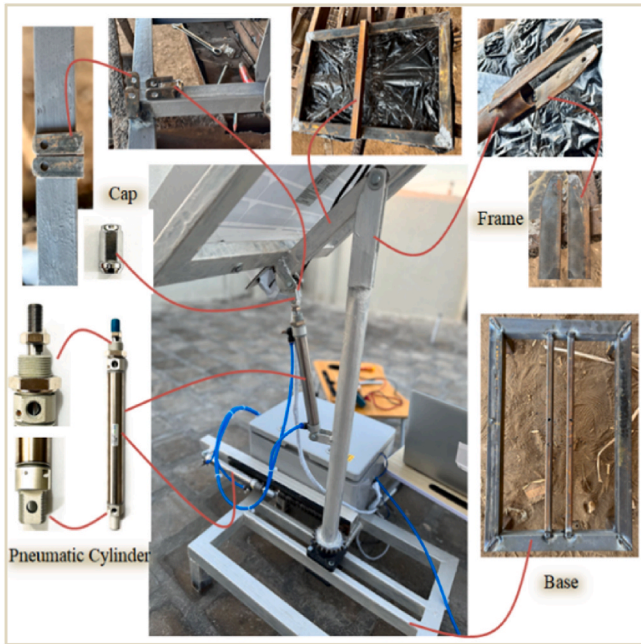


Fig. 7. The mechanical structure of the dual-axis solar tracker.

2.3. Implementation of the dual-axis tracking system

Fig. 6 shows the entire pneumatic solar tracking system setup. The system mainly consists of an electro-pneumatic system, a solar panel, the chassis (structure) that holds the panel, a control circuit, a power supply, and a monitoring device (PC). Sun angles are considered as input to the

Table 3

Control panel components.

Component	Quantity	Specification
Pneumatic valve	4	Model number: 3V210-08, coil voltage: 24VDC, valve body material: aluminum alloy, pressure range: 0.15–0.8mpa, port size: 1/4", structure: single solenoid 3 way 2 position normally closed.
Battery	2	ULTRAMAX NP9-12 12V 9AH 20HR
LDR	4	Max power dissipation: 200 mW, max voltage: 200V, peak wavelength: 600 nm, min. Resistance: 1.8 kΩ, max. Resistance: 4.5 kΩ.
Relay	4	Supply voltage: 3.75–6V, trigger current: 5 mA, current when the relay is active: ~70 mA (single), ~300 mA, relay maximum contact voltage: 250VAC, 30VDC, Relay max. current: 10A.
Arduino UNO	1	Microcontroller: ATmega328P – 8 bit AVR, operating voltage: 5V, recommended input voltage: 7–12V, input voltage limits: 6–20V, analog input pins: 6, digital i/o pins: 14 out of which 6 provide PWM output, dc current on i/o pins: 40 mA, dc current on 3.3V pin: 50 mA, Flash memory: 32 kB (0.5 kB is used for bootloader), SRAM: 2 kB, EEPROM: 2 kB, Freq.: 16 MHz
Resistor	4	Carbon resistance of 10 kΩ

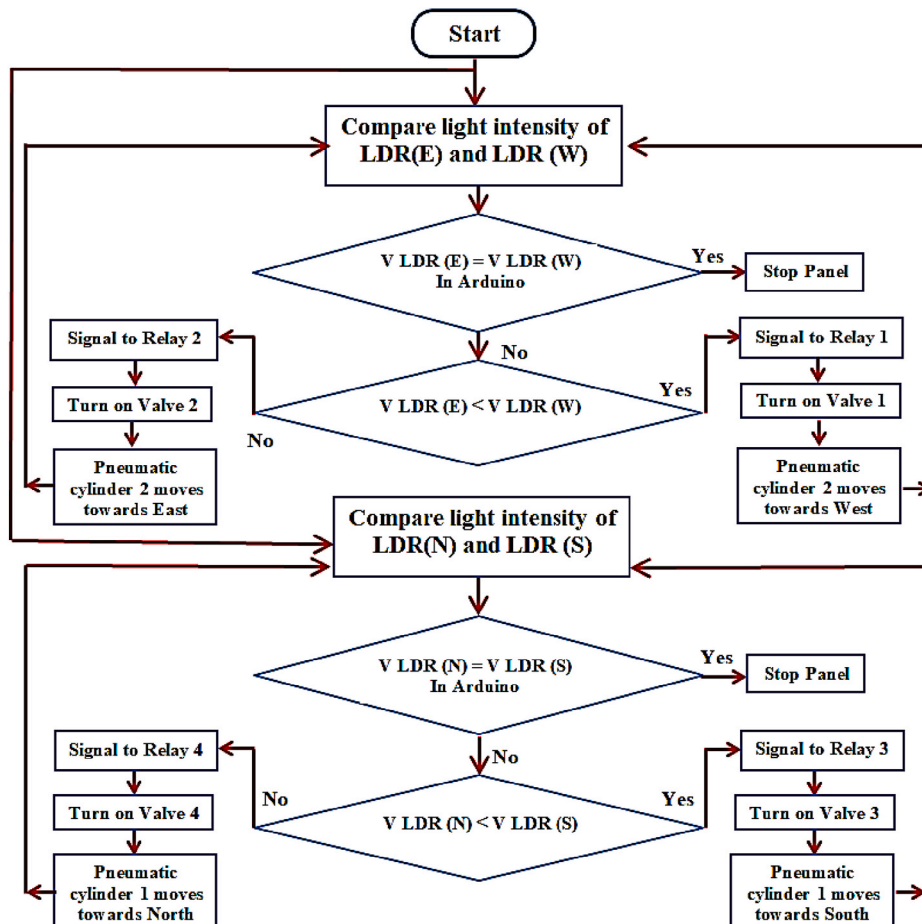


Fig. 8. Flowchart of the automatic operation of pneumatic dual-axis solar tracker.



Fig. 9. Control panel.

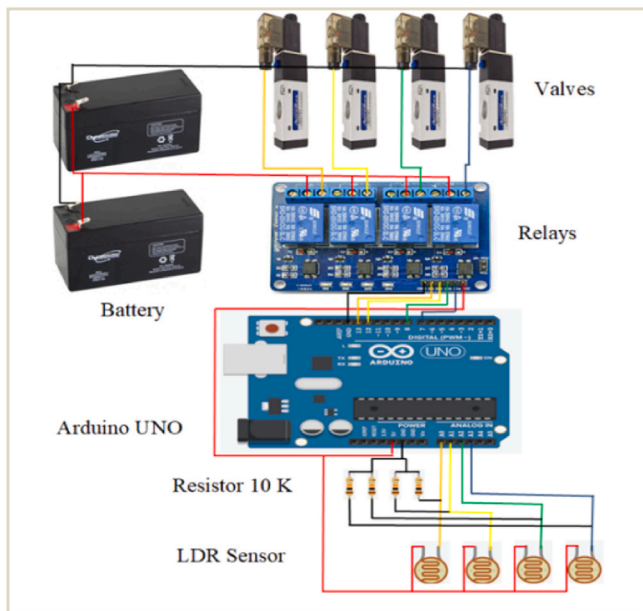


Fig. 10. Control of pneumatic cylinder's valves.

system, while output is the displacement of the pneumatic cylinders. The iron-formed structure consists of a frame that holds the PV panel. This frame is attached to a pillar via a free-moving joint to hold and facilitate vertical movement of the frame and panel assembly. The pillar is mounted from below on a base at the bottom by a rack gear, which, in turn, creates a semicircular movement of the tracking system. The pillar has a height of 692 mm and a circular cross-section with a diameter of 40 mm. The pneumatic system consists of two double-acting pneumatic

cylinders, four pneumatic solenoid valves, an air compressor, and a pressure regulator. Other details will be mentioned later.

Fig. 7 shows how the pneumatic actuators are mounted on the mechanical structure of the tracking system. The system has two pneumatic actuators to track the sun. The first actuator "1" tracks the linear vertical movement of the sun from south to north at different angles; between the pillar and the solar panel. The second actuator "2" tracks the horizontal movement of the sun from east to west. It is a linear motion that is converted into a circular motion to provide a maximum angle of up to 180° of precise tracking. An EU-M40W solar PV panel, with an area of 860 mm × 425 mm, thickness of 25 mm, and weight of 4 kg, was used in this design. The solar panel is mounted using screws on a rectangular iron frame with the same dimensions as the panel. The south-north rotation axis of the tracking system will be at the midpoint of the frame.

2.4. Automatic control panel

The pneumatic cylinder has two chambers. To operate the cylinder, the air inside the chamber forces the rod toward the other chamber. When controlling the position of the rod to follow a certain movement, the pressure inside the two chambers must be in certain proportions, and

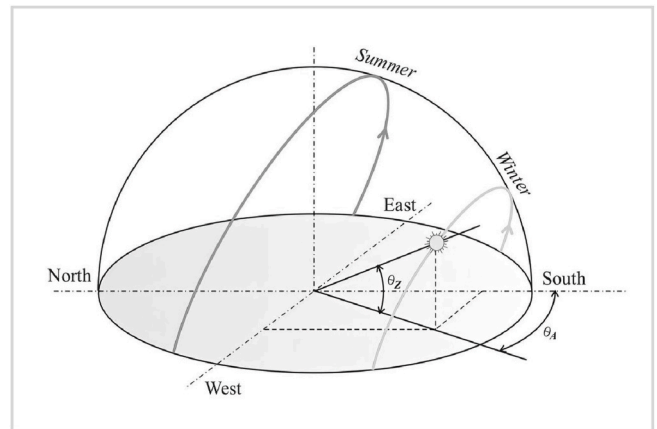


Fig. 11. Typical behavior for the sun's path.

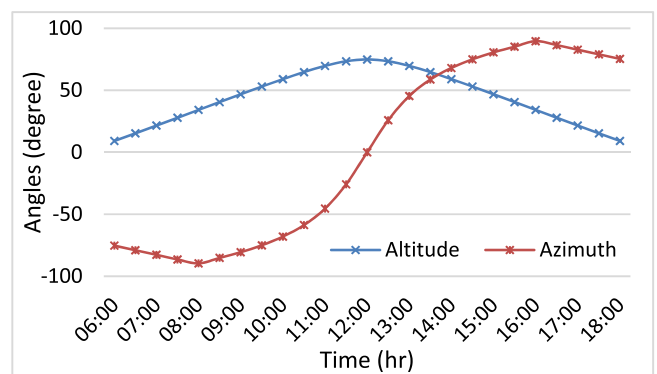


Fig. 12. Hourly distribution of Azimuth-Altitude angles during the experiment days.

Table 4
Suggested days for tests and related time information on May 2024.

Test no.	Test day	Movement type	Dawn	Sunrise	Culmination	Sunset	Daylight during
1	7 May	Fixed	04:45:13	05:11:24	12:00:22	18:49:48	13h, 38m, & 24s
2	8 May	Single-axis	04:44:19	05:10:33	12:00:19	18:50:31	13h, 39m, & 58s
3	9 May	Dual-axis	04:43:26	05:09:44	12:00:16	18:51:15	13h, 41m, & 31s

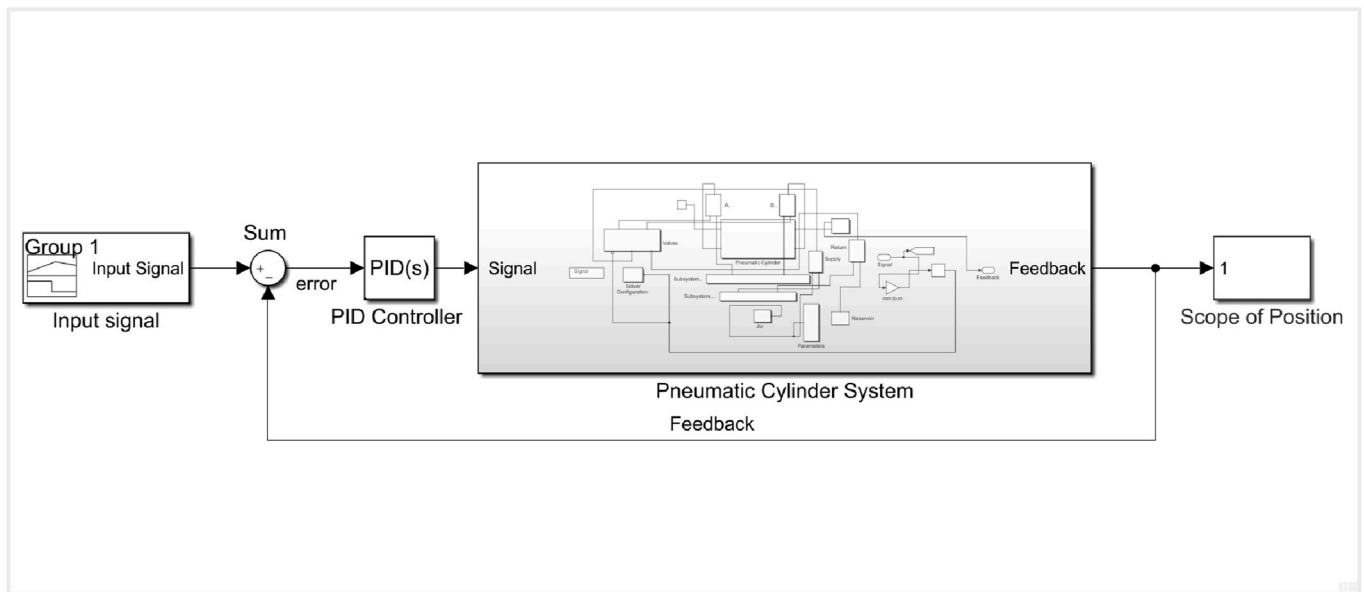


Fig. 13. Closed-loop control pneumatic system.

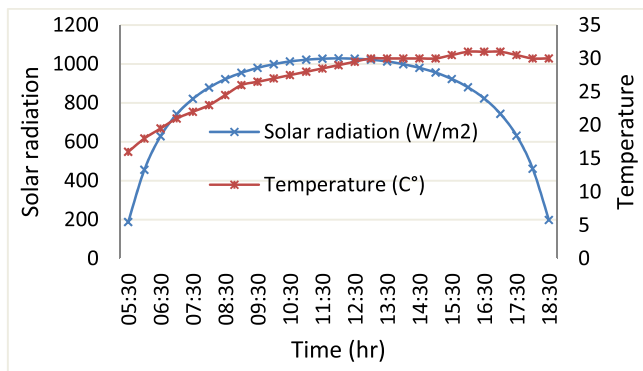


Fig. 14. Hourly distribution of solar radiation and ambient temperature.

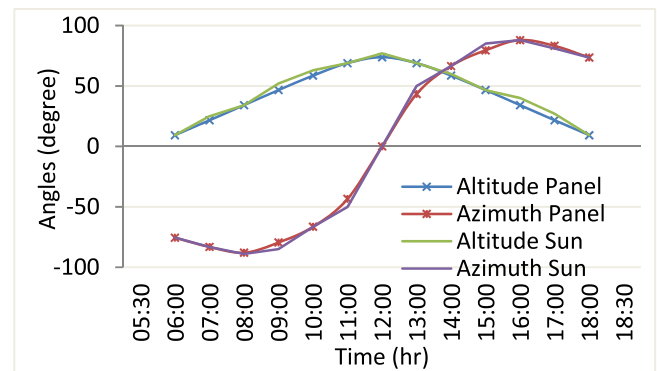


Fig. 16. Altitude and Azimuth scanning angles of the sun and PV panel.

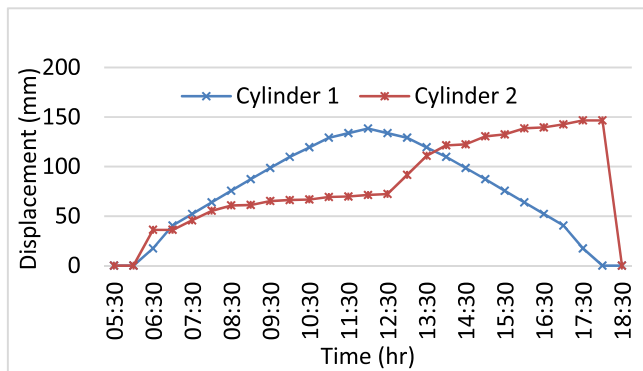


Fig. 15. Stroke displacement of the pneumatic cylinders.

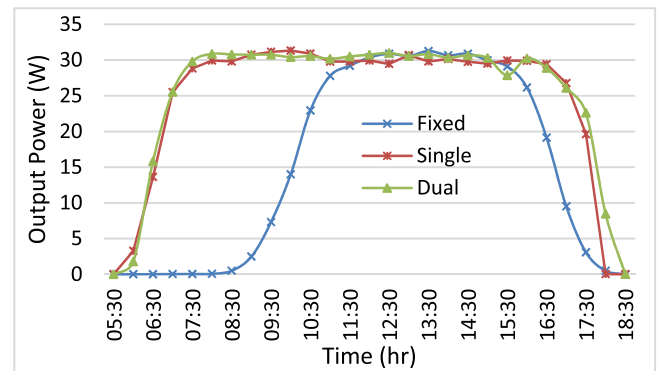


Fig. 17. Hourly distribution of PV panel output power under three cases of tracking.

the pressure should not be equal. The reason is that the air pressure upon entering the two chambers collides, and that is because two regions inside the two chambers differ in cross-sectional area. This is due to the presence of the rod as part of the cross-sectional area of one of the two chambers.

The pressure inside the pneumatic actuator must be maintained under the load condition, and the amount of air entering and leaving the

actuator must be regulated. Two valves must be positioned, one to regulate the pressure in chamber A and the other to control chamber B, to regulate the pressure within each pneumatic cylinder. In addition to establishing air throttles at the points where air enters the cylinder and exits the valve. To control the automatic mode of pneumatic actuators, four pneumatic valves, and four normally open relays must be used; one

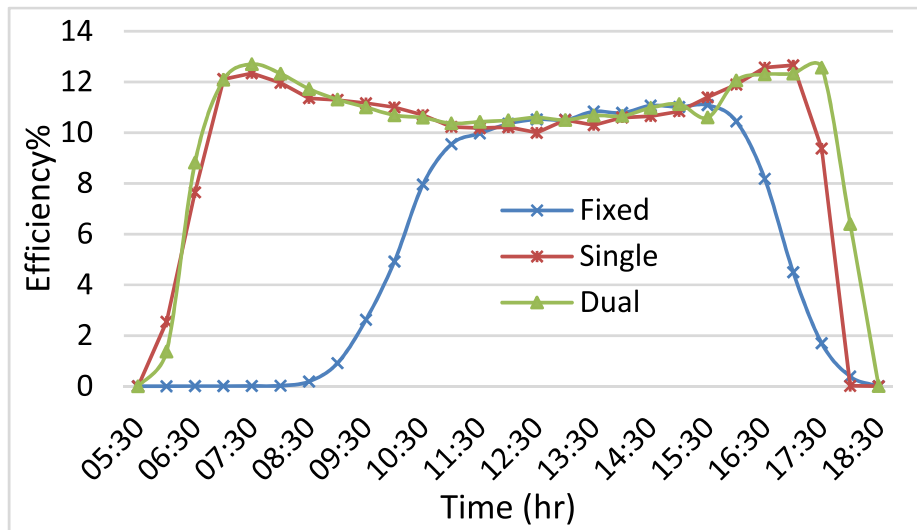


Fig. 18. Hourly distribution of PV panel efficiency under three cases of tracking.

Table 5

Statistical analysis data for the PV panel power and efficiency.

Data	Power			Efficiency		
	Fixed	Single-axis	Dual-axis	Fixed	Single-axis	Dual-axis
Standard error (%)	2.7498	0.5342	0.4594	0.9639	0.19162	0.1730
Standard deviation (%)	12.8979	2.5058	2.155	4.5211	0.8988	0.8115

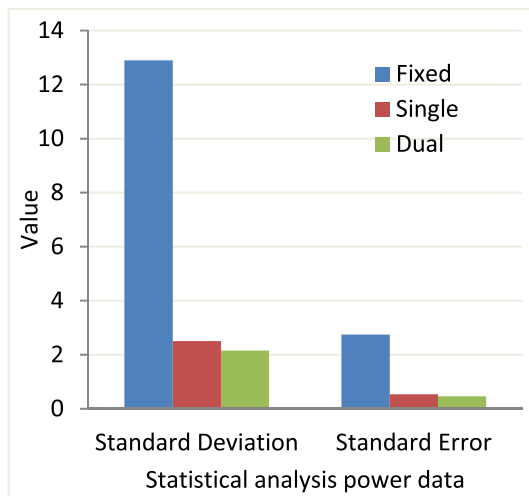


Fig. 19. Statistical analysis data of PV panel power.

relay must be assigned to each pneumatic valve.

Four LDR sensors as shown in Fig. 6 were employed to track the sun's beams. The LDR is a photoresistor, which is an optical sensor used to sense solar radiation. This unique kind of resistor varies with the intensity of light since it is composed of a high-resistance semiconductor that operates on the photoconductivity principle. The light's resistance decreases with increasing intensity. A plastic tube encircled each sensor, allowing it to detect any tilt of the sun and detect any variation in radiation, no matter how little. This provides more precision when positioning the solar panel perpendicular to the sun. The LDR sensors were at

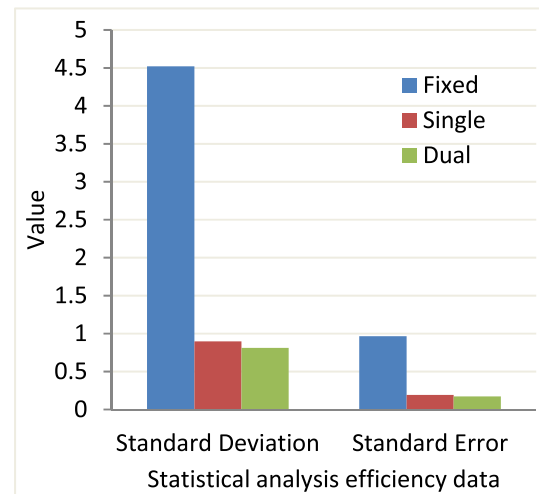


Fig. 20. Statistical analysis data of PV panel efficiency.

the same level as the PV solar panel, which received varying amounts of solar radiation based on the panel's rotation around the sun and converted it into electrical signals with varying values depending on the intensity of the radiation that is transmitted to the Arduino controller. In the controller, each pair of signals from two opposite LDR sensors is compared from south (S) to north (N) and from east (E) to west (W). After calculating the error rate between the two signals, the Arduino estimates the movement's direction based on the sun's movement. A specific relay that is connected to the related coil of the pneumatic valve receives a signal from the Arduino in response to this error. When the PV panel is oriented perpendicular to the sun's rays, the relay contact points revert to their open state since there is no comparison between east and west or south and north. How the LDR sensors work within the automatic control circuit can be seen in the flowchart in Fig. 8.

The specifications of the control panel components are listed in Table 3. Fig. 9 shows the control panel and its components. Fig. 10 shows the schematic of connecting the two-axis tracking components in the control panel using four pneumatic valves.

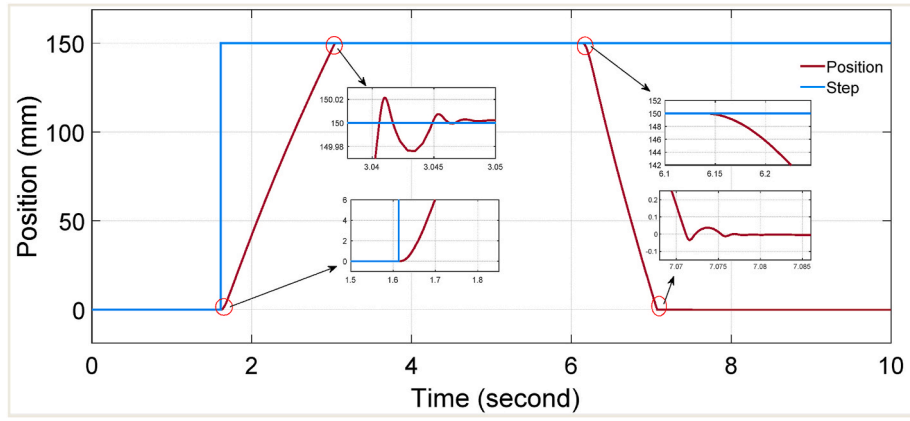


Fig. 21. Response to a step displacement (without PID controller).

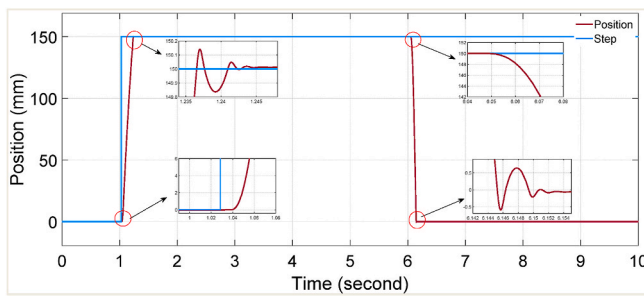


Fig. 22. Response to a step displacement (with PID controller).

2.5. Calculations of the Sun's position movement and panel efficiency

The experimental tests were conducted on the solar tracking system over three consecutive days, the 7th, 8th, and 9th of May, in the Iraqi city of Karbala, at latitude of 32.37570 and longitude of 44.03891. These days were chosen because they are similar to each other in terms of temperature distribution and are free of clouds. On the first day, the solar panel was fixed in the direction of latitude. On the second day, the system was moved in the azimuth direction with one degree of freedom from east to west, and the second axis was fixed in the direction of altitude. On the third day, the sun was tracked in two directions, azimuth and altitude with two degrees of freedom. Table 4 shows some information about the days proposed for conducting practical experiments. Fig. 11 shows an illustration of the sun's angles.

Pneumatic cylinder 1 is responsible for tracking the altitude angle (θ_z), which is the rotating angle of the orienting system in the vertical plane of space. This angle is computed using the below formula.

$$\sin(\theta_z) = \sin(\varphi) \sin(\delta) + \cos(\varphi) \cos(\delta) \cos(\omega) \quad (13)$$

Where $\theta_z = 90^\circ - \text{zenith angle of the sun}$, φ is the latitude ($\varphi = 32.37570^\circ$ for this example), ω is the hour angle ($15^\circ/\text{hour}$), where $\omega = 0$ at local noon, and δ is the solar declination and may be determined from Cooper's equation [49]:

$$\delta = 23.45 \sin \left[\frac{360}{365} (284 + N_d) \right] \quad (14)$$

Here, $N_d = 1$ represents one day out of the three days 7–9 May 2024.

On the other hand, pneumatic cylinder 2 is responsible for tracking the azimuth angle (θ_A), which represents the system's rotating angle in the horizontal plane. θ_A is computed as [50]:

$$\sin(\theta_A) = \frac{\cos(\delta) \sin(\omega)}{\cos(\theta_z)} \quad (15)$$

Fig. 12 shows the distribution of Azimuth-Altitude angles during the experiment days. From Eq. (7), the solar declination (δ) is found to be 17.5165° .

The efficiency of a PV panel (η) can be calculated based on the amount of solar radiation (G), the surface area of the panel (A_p), and its power output (P_o):

$$\eta(\%) = \frac{P_o}{G A_p} \times 100\% \quad (16)$$

2.6. Closed-loop control of pneumatic system

Stability guarantees that, despite changes in the system parameters, the system can achieve and sustain a steady state for a certain input. PID controller may be used for stability analysis, which aims to increase reaction speed with the least amount of error and overshoot by controlling the position of the pneumatic cylinder in the solar tracking system. Applying a position step signal to the pneumatic actuator will do this. Thus, as seen in Fig. 13, an effort was made to automatically adjust the PID controller's gain in the Matlab environment.

3. Results and discussion

3.1. System performance

The solar tracking device was experimentally tested over three days in a row on May 7, 8, and 9 in the Iraqi city of Karbala, at latitude 32.37570 and longitude 44.03891. These proposed days are fairly close

Table 6
System stability analysis for cylinder position (with and without controller).

Without PID controller				With PID controller			
+ Rise Time	+ Slew rate	Amplitude	Overshoot	+ Rise Time	+ Slew rate	Amplitude	Overshoot
1.096 (s)	108.464 (/s)	1.486×10^2 (mm)	0.505 %, -0.433 %	145.760 (ms)	819.030 (/s)	1.492×10^2 (mm)	0.505 %, 0.321 %
- Fall time	- Slew rate	Undershoot	Setting time	- Fall time	- Slew rate	Undershoot	Setting time
698.768 (ms)	-107.116(/s)	1.092 %, 0.505 %	1.617 (s)	61.016 (ms)	-1.957(/ms)	-0.301 %, 0.505 %	1.034 (s)

Table 7

Brief comparison of performance results with previous studies.

Study	Tracking type	Performance evaluation
[34]	Single	A semi-continuous sun tracker used around 34 % less energy than a continuous tracker.
[35]	Dual	Produced 27.75 % more power than fixed system, with $\pm 1.8^\circ$ position signal accuracy.
[36]	Dual	Two-axis-GPS sun tracker outperformed the static base & LDR base PV systems by 45.85 & 14.93 %, respectively.
[37]	Dual	Improved the PV efficiency by 28.5–42.9 % compared to the fixed system on sunny days & by 37 % on partly cloudy days.
[38]	Dual	Enhanced the initial fault coverage by 53 % within a brief simulation time of 2.5×10^{-8} s.
[39]	Single	The trackers based on astronomical programming & LDR performed better than the static PV system, with net gains of 31.8 % & 37 %, respectively.
[40]	Dual	Compared with & without the Fresnel lens, the output power increased by 16.03 %. When the tracker moved from E to W at 23° , the greatest power achieved with the lens was 15.90 W.
[41]	Dual	Extracted more electricity than the fixed-tilted tracker by 35.91 % in fixed locations with largely clear days, 38.72 % & 38.4 % in varied locations with clear days, & 60 % & 113.7 % in heading variation & movement circumstances.
[42]	Single	Produced up to 26.9 % more power than a fixed-tilted PV system in clear conditions
[43]	Dual	An accurate & efficient control system, making the hydraulic drive an essential part of producing efficient, clean energy.
[25]	Dual	18 % increase in power with an average power consumption of 100 pulses/hour in the steady-state. Each pulse has 100 ms.
[33]	Single	Dynamic overshoot response of 5.3 % & a steady-state error of 1.6 %.
[44]	Single	Under load & no-load situations, the average error is 2.5° & 1.6° , respectively. The system dynamic response was 1.25 % steady-state error & 2.08 % overshoot.
[27]	Dual	Increased total energy production by 37.6 % & actual energy production by 35.3 %, with an efficiency of 90 % compared to fixed panel system.
[46]	Dual	Compared to not using FL, rod position accuracy was improved by 12.35 %, & abrupt changes in input signals were handled better.
[26]	Dual	Achieve $\pm 80^\circ$ movement.
Present study	Dual	- The average efficiency of solar panel has been improved by 62.3 % in one axis when compared to the fixed panel, and by 66.4 % in the two axes of movement in Iraq weather conditions. - Under the PID controller, the pneumatic system responds with a rise time of 145.76 ms, a settling time of 1.034 s, an overshoot of 0.505 %, & a steady-state error of 0.00734 %.

in terms of the intensity of solar radiation and the ambient temperature. The experimental procedures began by calculating the distribution of solar radiation and ambient temperature during the day during the proposed three experimental days as shown in Fig. 14.

Cylinders 1 and 2's stroke displacement during the day is shown in Fig. 15. Between 5:30 and 18:30, the cylinders ran at their maximum displacement strokes of 138.38 and 146.6 mm, respectively. Because the sun's rays are not as perpendicular to the earth's surface at the start of spring (the current case study) as they are in the summer, it is noticed that the displacement of the cylinders did not exactly reach the design values of the stroke displacement previously mentioned in Table 2. The design values in Table 2 represent the maximum values based on the maximum displacement values of the stroke during the year.

Fig. 16 shows the scanning angles of the daylight movement for both the sun and the solar PV panel in the altitude and azimuth planes. The figure shows the degree of agreement between the panel's and the sun's

real motion. It is seen that, as a result of cylinder 1's movement, the panel starts to rise vertically and gradually at 5:30 till midday, reaching a maximum scanning angle of 73.8° (1.288 rad). After that, it descends until it resettles at sunset. Due to the action of gears powered by cylinder 2, the panel rotates horizontally at the same time, reaching a maximum angle of 175.9223° (3.07 rad) at the end of the day.

The output power (P_o) of the solar panel may be computed by multiplying its output voltage and current when it is loaded by a resistive load of 10Ω . The output power curves are displayed in Fig. 17, for three different cases: first, when the panel is stationary; second, when cylinder 2 drives the panel in one axis; and third, when the panel is driven by two cylinders in two axes. According to the figure, the average power output of the panel in the three scenarios was 15.1 W, 24.4 W, and 25 W, respectively.

The area of the panel ($0.67 \times 0.425 \text{ m}^2$), the radiation intensity (G) curve (Fig. 14), and the panel wattage curves (Fig. 17) that emerge from the panel can be used to draw the PV panel efficiency curves shown in Fig. 18 at the three cases of tracking according to Eq. (16).

It is noted from the figure that the mobile solar panel with two axes has somewhat better efficiency than those with one axis throughout the day, while its efficiency is much higher than the fixed panel. The efficiency increases with the increase in the value of solar radiation from the beginning of sunrise until approximately 7:30, after which efficiency begins to decrease until it reaches the lowest level at midday as a result of the temperature rise, which outweighs the increase resulting from the increase in solar radiation. After midday, the efficiency begins to rise until 17:30 as a result of the drop in temperature, and then it begins to drop as a result of the decline in solar radiation until it becomes zero at sunset. According to the figure, the movable PV panel with one axis has a 62.3 % greater average efficiency than the fixed panel, and the movable panel with two axes has a 66.4 % better average efficiency than the fixed panel. Several potential technical factors affect the dual-axis solar tracker efficiency, summarized as follows.

- High technical complexity
- Lots of moving parts make it more likely for components to fail
- Cloudy or overcast weather, wind load, and weather resistance
- Maximum load it needs to support
- Range of motion required for both axes
- Power consumption limitations
- Calibration procedures to ensure accurate tracking
- Mechanical tolerances
- Age of PV panels.
- Installation issues
- Geographical location.

3.2. Statistical analysis

All statistical analysis data related to the solar panel power and efficiency under different cases of tracking have been calculated and listed in Table 5. The bar chart visualization of data for statistical analysis of PV panel power and efficiency is shown in Figs. 19 and 20. Based on the results of statistical analysis, it can be concluded that the use of a dual-axis pneumatic solar tracking system increases energy production and enhances the overall performance of the system compared to a single-axis moving panel and fixed panel.

3.3. System stability

Stability, more specifically, guarantees that the system can achieve and sustain a stable state for a certain input. The stability of the pneumatic system should be analyzed to validate how efficiently the design meets the tracking requirements of the proposed solar system. The performance of the designed pneumatic system was evaluated by finding the response of the system to the displacement (position) step under two control conditions, without and with the PID controller. With the help of

the Matlab-Simscape toolbox, it was possible to automatically adjust the PID controller settings and get the appropriate control for each system. After carrying out the simulation, the optimal gain parameters in this procedure were determined to be $K_p = 65.102$, $K_i = 7.230$, and $K_d = 2.011$. The corresponding system step responses are shown in Figs. 21 and 22. The results of stability were recorded and listed in Table 6. The results demonstrate how the system reached a steady-state response with minimum error and overshoot and swiftly rising and settling step responses.

The results obtained in this study can be compared with the results of previous studies mentioned above and summarized in Table 7.

4. Conclusions

The study presents a simple and comprehensive design method for pneumatic actuators based on a number of variables, including tilt angle, actuator distance, and actuator power, to operate a dual-axis pneumatic tracking system specifically for the more extreme environment of the Iraqi city of Karbala. Analysis of the results allows some conclusions to be drawn.

- 1 The results demonstrated that the tracking system functions precisely, steadily, and smoothly when loaded under the greatest wind speed that is possible, which is around 5.5 m/s according to Iraqi climate statistics.
- 2 The proposed pneumatic system provides a simple and efficient way to actuate cylindrical valves in a nonlinear stroke position control system, which in turn drives the solar panel. It features high power output and fast movement speeds.
- 3 According to the experimental testing results, the average efficiency of the solar panel has been improved by 62.3 % in one axis when compared to the fixed panel, and by 66.4 % in the two axes of movement. These significant improvement rate figures demonstrate how well-suited and relevant the system is to Iraqi weather conditions in order to address the issue of electrical power outages.
- 4 The pneumatic system responded dynamically well under the PID controller with a rise time of 145.76 ms, a settling time of 1.034 s, an overshoot of 0.505 %, and a steady-state error of 0.00734 %.
- 5 The results showed that the proposed operational model, which can be applied to enhance rehabilitation equipment in the future, correctly captures the real behavior of the system.
- 6 The pneumatic actuator design procedures presented in this paper serve as a general guide that can be adopted in future research in the field of tracking systems driven by any other type of actuator.

CRedit authorship contribution statement

Mustafa A. Abdul-Hussein: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Jamal A.-K. Mohammed:** Writing – review & editing, Writing – original draft, Supervision, Software, Project administration, Investigation, Formal analysis, Conceptualization. **Wisam E. Abdul-Lateef:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare no conflict of interest.

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